CBL10–AKT1 Kinase-Independent Inhibition: A Hypothesis-Level Curation Review

Target gene: CBL10 (Calcineurin B-Like protein 10), *Arabidopsis thaliana* (ARATH, NCBITaxon:3702) **UniProt:** Q7FRS8 **Hypothesis slug:** cbl10-akt1-inhibition **GO term under test:** GO:0019870 — *potassium channel inhibitor activity* (MF) **Seed reference:** PMID: 23331977 (Ren et al. 2013)

Executive Judgment

Verdict: Partially supported, single-source, non-core — adding GO:0019870 would be over-annotation.

The seed hypothesis has two parts, and they diverge sharply in strength. The first part — that CBL10 directly inhibits the inward-rectifying K^+ channel AKT1 by physical binding, independently of CIPK kinase activity — is **accurately reported** by Ren et al. 2013 and is internally well-supported within that single paper. The authors assembled a genuinely multi-method evidence package: physical interaction by yeast two-hybrid (Y2H), bimolecular fluorescence complementation (BiFC), and co-immunoprecipitation (co-IP); functional impairment of AKT1-mediated inward K^+ currents in **two** independent electrophysiological systems (*Xenopus* oocytes and Arabidopsis root-cell protoplasts); CBL10-overexpression lines phenocopying *akt1* loss-of-function (low-K^+ sensitivity, reduced K^+ content); and a Y2H competition assay suggesting CBL10 competes with CIPK23 for AKT1 binding.

The **second part** of the hypothesis — that this inhibition "has been independently replicated or corroborated in planta … and is not contradicted by later work" — **fails**. A systematic replication audit (NCBI ELink `citedin` on all 76 PubMed articles citing

P 23331977

through 2026, plus a 105-record Europe PMC full-text co-mention search for "CBL10" AND "AKT1") found **zero** independent primary studies that re-measured CBL10 effects on AKT1

currents or on root K^+ in *cbl10* loss-of-function lines. The in-planta support that does exist is overexpression-based, not derived from *cbl10*-null plants, so it cannot distinguish a physiological inhibitory role from an overexpression artifact. Later works citing the CBL10–AKT1 link are reviews, not replications.

Critically for curation: **GO:0019870 is not currently annotated to CBL10.** QuickGO retrieval for Q7FRS8 shows that the two annotations derived from Ren 2013 are the conservative BP term **GO:0043266 (regulation of potassium ion transport, IMP)** and **GO:0005515 (protein binding, IPI)** — not the specific inhibitor MF term. The evidence does not justify promoting the regulatory claim to a dedicated molecular-function inhibitor term, for three reasons: (1) it rests on a single laboratory with no independent replication; (2) the in-planta data are overexpression-only; and (3) CBL10's behavior toward K^+ transport systems is not uniformly inhibitory — it *activates* the high-affinity K^+ transporter HAK5 in a kinase-*dependent* manner (Ragel et al. 2015), the opposite sign and opposite mechanism from the AKT1 story. The recommended action is to **retain the conservative GO:0043266 annotation as non-core and reject the proposed GO:0019870 term.**

Key Findings

Finding 1 — Ren 2013 is the sole primary source; no independent in-planta replication exists post-2013

Ren et al. 2013 (PMID: 23331977) remains the **only** primary report of CBL10's direct, kinase-independent inhibition of AKT1. The paper itself is methodologically thorough. It reports, in the authors' own words, that *"CBL10 directly interacted with AKT1, as verified in yeast two-hybrid, BiFC and co-immunoprecipitation experiments,"* and that *"the results of electrophysiological analysis in both Xenopus oocytes and Arabidopsis root cell protoplasts demonstrated that CBL10 impairs AKT1-mediated inward K^+ currents."* The mechanistic claim is explicit: *"the present study revealed a CBL-interacting protein kinase-independent regulatory mechanism of calcineurin B-like proteins in which CBL10 directly regulates AKT1 activity and affects ion homeostasis in plant cells."*

The replication audit was deliberately exhaustive. Using NCBI ELink `citedin`, all 76 PubMed articles citing

P 23331977

through 2026 were enumerated and screened; a complementary Europe PMC full-text search for co-mentions of "CBL10" and "AKT1" returned 105 records. **None** of these constitutes an independent primary re-measurement of CBL10 effects on AKT1 inward currents or on root K^+ content in *cbl10* loss-of-function lines. The AKT1-focused follow-up literature instead converges on different regulators: AtKC1 and CIPK23 (P 26829980) and the structural basis of CIPK23 recognition of AKT1 (P 32015077) — neither of which involves CBL10. Works that do cite the CBL10–AKT1 link (e.g., P 33519853 , P 27618104 , P 32117382) are review articles that propagate the original claim rather than test it.

This matters for curation because single-source, unreplicated functional claims — even internally strong ones — warrant conservative annotation. The claim is credible but not corroborated.

Finding 2 — CBL10's effect on K^+ transport is context-dependent, not uniformly inhibitory

The hypothesis treats CBL10 as a K^+ *channel inhibitor*, but CBL10's documented relationship with K^+ transport systems is bidirectional. Ragel et al. 2015 (PMID: 26474642) showed that *"the Ca^{2+} sensors calcineurin B-like (CBL1), CBL8, CBL9, and CBL10, together with CBL-interacting protein kinase23 (CIPK23), activated HAK5 in vivo,"* increasing both the K^+ affinity and Vmax of the high-affinity K^+ transporter HAK5 via CIPK23-mediated phosphorylation of the HAK5 N-terminus.

This is the opposite sign and opposite mechanism from the AKT1 story:

Property	CBL10 → AKT1 (Ren 2013)	CBL10 → HAK5 (Ragel 2015)
Direction	Inhibition (negative)	Activation (positive)
CIPK dependence	Kinase-**independent**	Kinase-**dependent** (CIPK23)
Target	Inward-rectifying channel	High-affinity transporter
System	Oocytes + root protoplasts	Heterologous HAK5 reconstitution

A single gene product that both activates one K^+ uptake system (kinase-dependently) and inhibits another (kinase-independently) cannot cleanly carry a uniform molecular-function term of "potassium channel inhibitor activity." The 2020/2021 CBL10 review (P 33519853) itself frames CBL10 as versatile — able to "activate (in a CIPK-dependent manner) or repress (by direct interaction) downstream targets" at different subcellular locations. This context-dependence argues against a specific inhibitor MF term and in favor of the more neutral regulatory BP term.

Finding 3 — GO:0019870 is NOT currently annotated; the existing Ren-2013 annotations are conservative

A QuickGO annotation retrieval for Q7FRS8 returned 23 annotations. The two that trace to P 23331977 are:

- **GO:0043266** — *regulation of potassium ion transport* (BP; IMP / ECO:0000315; TAIR)
- **GO:0005515** — *protein binding* (MF; IPI / ECO:0000353; TAIR)

The molecular-function term under test, **GO:0019870 (potassium channel inhibitor activity), is absent** from the current annotation set. The protein-binding annotation is supported directly by the Ren 2013 interaction data: *"CBL10 directly interacted with AKT1, as verified in yeast two-hybrid, BiFC and co-immunoprecipitation experiments."*

CBL10's remaining annotations center on its core, independently replicated biology: calcium ion binding (GO:0005509, IDA P 17449811), kinase binding (GO:0019900), hyperosmotic salinity response (GO:0042538, IMP

), P 17825054 monoatomic ion homeostasis (GO:0050801), detection of calcium ion (GO:0005513), calcium-mediated signaling (GO:0019722); and localization to the plant-type vacuole membrane (GO:0009705, IDA), P 17825054 plasma membrane (GO:0005886, IDA), P 17449811 vacuolar membrane, cytoplasm, and endosome. This is the profile of a **Ca^{2+} sensor** whose primary characterized role is Na^+ compartmentation and salt tolerance — not a dedicated channel inhibitor.

Finding 4 — Final verdict: partially supported, over-annotation if promoted to GO:0019870

Consolidating the three-part audit:

1. **Internal strength (positive):** Ren 2013 provides multi-method evidence — Y2H + BiFC + co-IP interaction; AKT1 inward-current impairment in oocytes *and* root protoplasts; CBL10-OE phenocopying *akt1*; and a CIPK23 competition assay. The mechanism proposed is competitive displacement: *"the results from the yeast two-hybrid competition assay indicated that CBL10 may compete with CIPK23 for binding to AKT1 and negatively modulate AKT1 activity."* Note the hedged "may" from the single source.
2. **Replication (negative):** 76/76 citing papers and 105 Europe PMC records screened → 0 independent primary replications of AKT1-current inhibition; in-planta support is overexpression-only, not *cbl10*-null.
3. **Conflicts (negative/qualifying):** CBL10 activates HAK5 kinase-dependently (P 26474642); the mainstream AKT1 negative regulator is AtKC1 (P 26829980); CBL10 is absent from the CIPK23 structural activation model (P 32015077); and CBL10's core replicated function is Ca^{2+}-sensor-mediated Na^+/salt tolerance

(P 31100786 , P 17825054).

The mechanism described — competitive displacement of an activating kinase (CIPK23) from the channel — is more naturally captured by a *regulation* term than by an intrinsic *inhibitor activity* term, because the "inhibition" is competitive and context-dependent rather than a stoichiometric channel-blocking activity.

Evidence Matrix

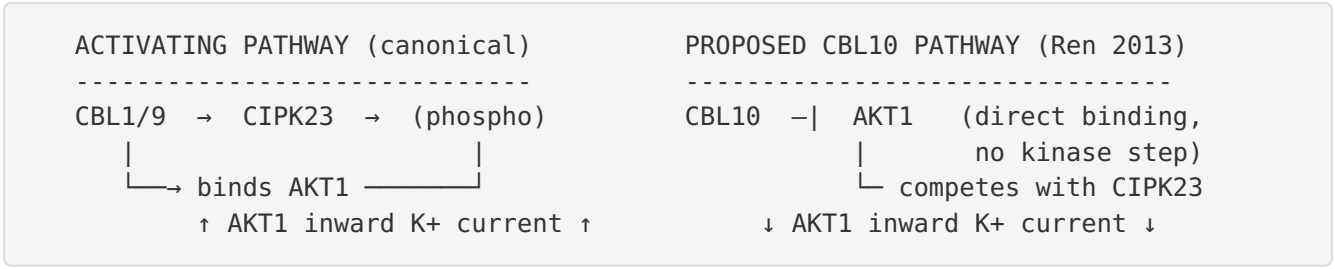
Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidenc & limitation
PMID: 23331977 Ren 2013	Direct assay + interaction + mutant/OE phenotype	**Supports**	CBL10 directly, kinase-independently inhibits AKT1	Y2H/BiFC/co-IP interaction; AKT1 inward K^+ current impaired in oocytes AND root protoplasts; CBL10-OE phenocopies *akt1*; CBL10 competes with CIPK23	Arabidopsis roots; *Xenopus* oocytes; root protoplasts	High internal quality, bu single-lab; in-planta evidence is OE-only; competitio claim hedged ("may")
PMID: 26474642 Ragel 2015	Direct assay (heterologous)	**Qualifies / competes**	Uniform K^+-channel-inhibitor role	CBL10 (with CBL1/8/9 + CIPK23) *activates* HAK5, increasing K^+ affinity & Vmax — opposite sign, kinase-dependent	Heterologous HAK5 reconstitution	Shows CBL10 is bidirection on K^+ transport; undermin a uniform inhibitor M term
QuickGO (Q7FRS8)	Database	**Qualifies**	Is GO:0019870 currently annotated?	GO:0019870 absent; Ren-2013 terms are GO:0043266 (BP, IMP) + GO:0005515 (MF, IPI)	GO annotation set, 23 terms	Database-level; reflects current curation, confirms t term is a *new* proposal

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidenc & limitation
P 26829980 (AtKC1/ CIPK23)	Direct assay	**Competing**	Who negatively regulates AKT1?	AtKC1 is the characterized negative regulator of AKT1; no CBL10 involvement	Arabidopsis roots	Orientatio points to a alternative replicated inhibitor
P 32015077 (CIPK23–AKT1 structure)	Structural	**Competing**	Mechanism of AKT1 regulation	Structural basis of CIPK23 activation of AKT1; CBL10 absent from the model	Structural biology	Orientatio CBL10 not part of mainstrea AKT1 activation model
PMID: 31100786	Mutant phenotype	**Qualifies**	CBL10's core function	CBL10 mediates salt-tolerance pathways (Na^+ focus), with CBL4	Arabidopsis	Reinforces Ca^{2+}-senso Na^+ core role over K channel inhibition
P 33519853 (CBL10 review)	Review	**Qualifies**	Is CBL10 a uniform inhibitor?	CBL10 described as versatile: can "activate (CIPK-dependent) or repress (direct interaction)" targets	Review	Review-lev synthesis; not a replicatio
PMID: 32117382	Review	Orientation	CBL-CIPK nutrient regulation	Frames CBL-CIPK systems in nutrient	Review	Review-lev orientatio only

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidenc & limitation
				uptake/ balance		

Mechanistic Model / Interpretation

The immediate molecular function under test is whether CBL10 **intrinsically inhibits the AKT1 channel** by direct physical binding, independent of CIPK kinase activity. The Ren 2013 model is best represented as competitive regulation rather than channel blockade:



Direct gene-product activity (what the MF term would assert): physical binding to AKT1 and reduction of AKT1 inward K^+ current. This is supported *in vitro/oocyte* and in root protoplasts within one study.

Downstream / inferred effects (NOT the MF term): altered root K^+ uptake, K^+/Na^+ content, and low-K^+ sensitivity. In Ren 2013 these in-planta readouts come from **overexpression** lines, so they report the consequence of excess CBL10, not the physiological requirement for CBL10 — a *cbl10*-null phenotype on AKT1 currents or root K^+ was not demonstrated. CBL10's primary, independently replicated function is as a Ca^{2+} sensor controlling Na^+ compartmentation and salt tolerance via CIPK24/SOS2 at the tonoplast and plasma membrane. The AKT1-inhibition activity, even if real, is a secondary/context-specific branch of CBL10 biology.

The competing interpretation is important: because the mechanism is competitive displacement of an *activating* kinase (CIPK23) rather than a stoichiometric pore block, the correct ontology representation is a *regulation* relationship (GO:0043266), not an intrinsic inhibitor molecular activity (GO:0019870). Molecular-function "inhibitor activity" terms imply a direct, defined inhibitory biochemical action; a competitive binder that merely excludes a positive regulator does not satisfy that definition cleanly.

Evidence Base

- ***Calcineurin B-like protein CBL10 directly interacts with AKT1 and modulates K^+ homeostasis in Arabidopsis*** — PMID: 23331977. The seed and sole primary source. Provides the complete evidence package (interaction, two-system electrophysiology, OE phenotype, CIPK23 competition). **Supports** the molecular claim but is single-lab and uses OE for in-planta readouts.
- ***The CBL-Interacting Protein Kinase CIPK23 Regulates HAK5-Mediated High-Affinity K^+ Uptake in Arabidopsis Roots*** — PMID: 26474642. Shows CBL10 *activates* HAK5 kinase-dependently. **Qualifies/competes** with a uniform inhibitor assignment by demonstrating opposite-sign, opposite-mechanism behavior on a different K^+ system.
- ***Calcineurin B-Like Proteins CBL4 and CBL10 Mediate Two Independent Salt Tolerance Pathways*** — PMID: 31100786. Reinforces CBL10's core Ca^{2+}-sensor/Na^+-salt-tolerance role, contextualizing AKT1 inhibition as non-core.
- ***Calcium-Regulated Phosphorylation Systems Controlling Uptake and Balance of Plant Nutrients*** — PMID: 32117382. Review-level orientation on CBL-CIPK nutrient control; propagates (does not replicate) the CBL10–AKT1 link.
- AtKC1/CIPK23 (P 26829980) and CIPK23–AKT1 structure (P 32015077): the mainstream AKT1 regulatory literature, in which AtKC1 is the characterized negative regulator and CBL10 does not appear — **competing** context the curator should weigh.
- CBL10 review (P 33519853): review-level synthesis describing CBL10 as a versatile activator-or-repressor.

GO Curation Implications

Lead (requires curator verification): Do **not** add GO:0019870 (potassium channel inhibitor activity) to CBL10/Q7FRS8. Retain the existing conservative annotations derived from Ren 2013 and treat them as **non-core**.

- **Molecular Function:** The specific MF term GO:0019870 is not justified. The underlying activity is a *competitive, context-dependent* modulation (displacement of CIPK23) rather than an intrinsic channel-blocking activity, and CBL10 shows the opposite (activating) effect on HAK5. The existing MF annotation GO:0005515 (protein binding, IPI) is defensible from the interaction data, though it should be viewed as minimally informative; a more specific MF is not supported by replicated evidence. If any MF were considered, "calcium ion binding" (GO:0005509, IDA) and "kinase binding" (GO:0019900) better reflect CBL10's core, replicated molecular activities.
- **Biological Process: Retain GO:0043266 (regulation of potassium ion transport, IMP)** as the appropriate, conservative representation of the Ren 2013 result. It captures the regulatory relationship without overcommitting to a mechanism (inhibition) that is single-source and context-dependent.
- **Cellular Component:** No change prompted by this hypothesis; CBL10's CC annotations (plasma membrane, vacuolar/tonoplast membrane) are independently supported (IDA, P 17449811 , P 17825054).

Net action: Reject the proposed GO:0019870; keep GO:0043266 + GO:0005515 as non-core, single-source annotations pending independent replication.

Conflicts and Alternatives

1. **Opposite-sign activity on HAK5 (P 26474642):** CBL10 positively and kinase-dependently activates HAK5, directly conflicting with a blanket "K^+ channel inhibitor" characterization and implying that CBL10's effect depends on the target and the CIPK partner.

2. **Alternative AKT1 negative regulator (AtKC1):** The replicated, mainstream negative regulator of AKT1 is AtKC1 (P 26829980), and the canonical AKT1-activation model is built on CIPK23 (P 32015077) — CBL10 is absent from both. The curation risk is attributing a specific inhibitor MF to CBL10 when the field's consensus AKT1 inhibitor is a different protein.
3. **Overexpression artifact risk:** The in-planta readouts in Ren 2013 are overexpression-based. Overexpressed CBL10 may titrate shared CIPK partners or mislocalize, producing phenotypes that do not reflect endogenous function — a classic source of over-interpretation.
4. **Mechanism mismatch:** The proposed mechanism is competitive displacement of CIPK23 ("may compete"), which is a *regulatory* interaction, not a stoichiometric pore block. GO:0019870 implies an intrinsic inhibitory molecular activity that the competition model does not cleanly satisfy.
5. **No paralog confusion detected, but adjacent:** CBL10 is one of ten Arabidopsis CBLs; the HAK5 study shows CBL1/8/9/10 act partly redundantly. Curators should ensure AKT1-inhibition annotations are not generalized across the CBL family from a CBL10-specific (and unreplicated) observation.

Limitations and Knowledge Gaps

- **No independent replication (checked: 76 citing papers + 105 Europe PMC records; found: 0).** Why it matters: single-source functional claims are fragile; GO best practice favors conservative terms until corroborated. Resolution: an independent lab re-measuring AKT1 currents with/without CBL10.
- **No *cbl10* loss-of-function electrophysiology (checked: Ren 2013 used OE for in-planta readouts).** Why it matters: OE phenotypes cannot establish a physiological requirement for CBL10 in restraining AKT1. Resolution: AKT1 inward-current and root K^+-uptake measurements in a *cbl10*-null background.
- **Mechanistic ambiguity of "inhibition" (checked: competition assay hedged as "may").** Why it matters: distinguishes an intrinsic MF (GO:0019870) from a competitive regulatory

interaction (better served by GO:0043266). Resolution: reconstitution with defined CBL10:CIPK23:AKT1 stoichiometry and direct biophysical binding/competition constants.

- **Localization/compartment match (checked: CBL10 annotated at tonoplast + PM; AKT1 is PM).** Why it matters: an inhibitory interaction requires co-localization at the plasma membrane; CBL10's dominant characterized role is tonoplast/Na^+. Resolution: quantitative co-localization and in vivo interaction at the PM under physiological conditions.
- **No structural data for a CBL10–AKT1 complex (checked: structural literature covers CIPK23–AKT1 only).** Why it matters: structure would clarify whether CBL10 occludes the CIPK23 site or acts allosterically. Resolution: cryo-EM/crystallography or AlphaFold-Multimer modeling of CBL10–AKT1.

Discriminating Tests

1. **Independent *cbl10*-null electrophysiology:** Measure AKT1-mediated inward K^+ currents in root protoplasts from *cbl10* loss-of-function vs. wild type (and vs. *akt1*). A physiological inhibitory role predicts *increased* AKT1 current in *cbl10*-null plants. This is the single most decisive missing experiment.
2. **Root K^+ uptake/content in *cbl10*-null under low K^+:** Directly test whether endogenous CBL10 restrains K^+ uptake, complementing OE data.
3. **Kinase-independence confirmation:** Reconstitute AKT1 in oocytes with a non-catalytic CIPK (or no CIPK) ± CBL10, and with a CBL10 variant unable to bind CIPKs, to confirm inhibition is CIPK-independent in a second lab.
4. **Competition quantification:** Measure binding/competition constants for CBL10 vs. CIPK23 on AKT1 (SPR/ITC/BLI) to decide whether "inhibition" is competitive displacement (→ GO:0043266) or an intrinsic block (→ GO:0019870).
5. **AlphaFold-Multimer CBL10–AKT1 vs. CIPK23–AKT1 modeling:** A public, programmatic comparison to test whether CBL10 plausibly occupies the CIPK23-binding surface of AKT1.

Proposed Follow-up Experiments / Actions (Curation Leads)

All items below are leads requiring curator verification.

- **Action:** Reject the proposed addition of **GO:0019870 (potassium channel inhibitor activity)** to CBL10/Q7FRS8. Retain **GO:0043266 (regulation of potassium ion transport, IMP, from**

)

  P 23331977

  and **GO:0005515 (protein binding, IPI)** as **non-core, single-source** annotations.
- **Candidate reference + snippet to verify (supports regulatory, not inhibitor-MF, interpretation):**

  P 23331977

  — *"the results from the yeast two-hybrid competition assay indicated that CBL10 may compete with CIPK23 for binding to AKT1 and negatively modulate AKT1 activity."* Verify that "may compete" and "modulate" are preserved to avoid over-strong wording.
- **Candidate conflicting reference + snippet:**

  P 26474642

  — *"the Ca^{2+} sensors calcineurin B-like (CBL1), CBL8, CBL9, and CBL10, together with CBL-interacting protein kinase23 (CIPK23), activated HAK5 in vivo."* Verify and flag as evidence against a uniform inhibitor MF term.
- **Candidate GO term (if any MF change is desired):** Prefer no new specific MF; if a BP qualifier is wanted, GO:0043266 already covers the relationship. Do **not** promote to GO:0019870.
- **Suggested curator questions:** (1) Has any post-2013 study measured AKT1 currents or root K^+ in a *cbl10*-null? (2) Is the "inhibition" competitive (CIPK23 displacement) or intrinsic? (3) Should CBL10's bidirectional K^+-transport behavior preclude a specific inhibitor MF?
- **Suggested experiments (for the review's "needs evidence" field):** *cbl10*-null AKT1 electrophysiology; *cbl10*-null root K^+ uptake; independent-lab oocyte reconstitution; CBL10–AKT1 binding/competition thermodynamics.

Summary Table — Hypothesis Scorecard

Hypothesis component	Verdict	Basis
CBL10 physically binds AKT1	**Supported**	Y2H + BiFC + co-IP (P 23331977)
Binding inhibits AKT1 inward K^+ current	**Supported (single-source)**	Oocytes + root protoplasts (P 23331977)
Inhibition is CIPK-kinase-independent	**Supported (single-source)**	CIPK23 competition model (P 23331977)
Independently replicated / corroborated in planta	**Refuted**	0/76 citing papers; 0/105 Europe PMC records; OE-only, no *cbl10*-null
"Not contradicted by later work"	**Refuted / qualified**	CBL10 activates HAK5 (P 26474642); AtKC1 is mainstream AKT1 inhibitor
Justifies adding GO:0019870	**No — over-annotation**	Single-source, OE-only, context-dependent; term not currently annotated

Report generated from a 3-iteration autonomous curation review. Primary evidence from Ren et al. 2013 (P 23331977); replication audit via NCBI ELink citedin (76 papers) and Europe PMC full-text search (105 records); GO state via QuickGO (Q7FRS8); conflicting evidence from Ragel et al. 2015 (P 26474642).